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Beyond the AI Divide

A Simple Approach to Identifying Global and Local Overperformers in AI Preparedness

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Abstract

This paper examines global disparities in artificial intelligence preparedness, using the 2023 Artificial Intelligence Preparedness Index developed by the International Monetary Fund alongside the multidimensional Economic Complexity Index. The proposed methodology identifies both global and local overperformers by comparing actual artificial intelligence readiness scores to predictions based on economic complexity, offering a comprehensive assessment of national artificial intelligence capabilities. The findings highlight the varying significance of regulation and ethics frameworks, digital infrastructure, as well as human capital and labor market development in driving artificial intelligence overperformance across different income levels. Through case studies, including Singapore, Northern Europe, Malaysia, Kazakhstan, Ghana, Rwanda, and emerging demographic giants like China and India, the analysis illustrates how even resource-constrained nations can achieve substantial artificial intelligence advancements through strategic investments and coherent policies. The study underscores the need for offering actionable insights to foster peer learning and knowledge-sharing among countries. It concludes with recommendations for improving artificial intelligence preparedness metrics and calls for future research to incorporate cognitive and cultural dimensions into readiness frameworks.

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Beyond the AI Divide:
A Simple Approach to Identifying Global and Local Overperformers in AI Preparedness

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I- Introduction

The transformative potential of artificial intelligence (AI) is reshaping economies, societies, and global competitiveness (Brynjolfsson and Unger, 2023), making its adoption a critical priority for nations worldwide.[2] The most prominent accounting and consulting firms claim that AI is projected to contribute significantly to the global economy, with estimates ranging from US$2.6 trillion to US$4.4 trillion annually (Chui et al., 2023), to US$15.7 trillion by 2030, driven by productivity and consumption gains (PwC, 2017). Some recent NBER publications[3] highlight that AI has the potential to significantly boost productivity and economic growth by automating tasks and driving innovation. However, the realization of these gains depends on complementary policies, institutions, and societal adaptation to ensure equitable and sustainable outcomes (Aghion, Jones, and Jones, 2019; Trammell and Korinek, 2023; Acemoglu, 2024). To summarize, AI, and particularly generative AI, has the potential to boost economic growth and productivity but poses risks of labor market disruptions, including premature de-professionalization—where skilled jobs are replaced or diminished before workers can fully capitalize on their expertise—and challenges for developing economies, such as reduced opportunities for high-skill job creation, increased inequality, and difficulties in integrating AI technologies into less-developed industrial and educational systems (Liu, 2024).

Such substantial socio-economic impacts underscore the critical importance for countries to adopt and integrate AI technologies to remain competitive and keep a sovereign public policy.[4] However, the benefits of AI are not uniformly distributed. The 2023 Government AI Preparedness Index (AIPI) from the International Monetary Fund (IMF) reveals significant disparities in AI preparedness and readiness among nations, highlighting a growing divide between those equipped to leverage AI and those at risk of falling behind (Cazzaniga et al., 2024).[5] Related to this, Liu and Wang (2024) highlight that middle-income countries exhibit unexpectedly high engagement with generative AI tools relative to their economic scale, while low-income economies are notably underrepresented. In this context, it becomes imperative to identify effective strategies for enhancing a country's AI readiness. This paper aims to explore such strategies, providing a framework for nations to efficiently bolster their AI capabilities and ensure inclusive growth in the evolving digital landscape.

The methodology outlined in this paper is straightforward: it evaluates AI preparedness at the country level using the AIPI and compares it to each country's assessed level of economic complexity. This approach identifies countries that outperform in AI preparedness and readiness relative to their

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[2] The UN Secretary-General's High-level Advisory Body on Artificial Intelligence (HLAB-AI) has published the final report "*Governing AI for Humanity*," advocating for the establishment of a globally inclusive and distributed AI institutional framework rooted in international cooperation. The report emphasizes the need for collaboration among governments and stakeholders to ensure AI promotes development while safeguarding human rights (UN HLAB-AI, 2024).

[3] National Bureau of Economic Research (NBER) publications, including working papers, are highly regarded as they are produced by leading economists, often serve as precursors to top peer-reviewed journal articles, and are widely cited in academic and policy discussions.

[4] "The late 2022 launch of GPT-3.5, OpenAI's generative Artificial Intelligence (AI) chatbot, sparked a media frenzy about the onrushing evolution of AI and what it portends for the human race and the planet at large. During the recently concluded Global Digital Summit, World Bank President Ajay Banga observed that 'even people who are relatively naive about AI are rushing toward AI, because of the fear of being left behind.'" [https://accountability.worldbank.org/en/news/2024/Developing-AI-for-development].

[5] See https://www.imf.org/external/datamapper/datasets/AIPI.

economic complexity for a given income level, aiming to uncover the factors driving their overperformance. Economic complexity refers to the knowledge intensity of an economy, captured by the diversity and sophistication of the goods and services it produces. The Economic Complexity Index (ECI), initially developed by Hidalgo and Hausmann (2009), measures this complexity by analyzing a country's export mix, reflecting its capabilities to produce and export diverse, high-value products. More recently, the ECI expanded to technology, based on patents data, and research, based on research papers (Stojkoski, Koch, and Hidalgo, 2023), since trade data can miss key information about innovative activities.

Using this methodology, we categorize distinct groups of AI overperformers based on their Economic Complexity Index (ECI) and their preparedness and readiness levels. Notably, overperformers are not simply the countries with the highest AIPI scores but those that (i) meet a minimum AIPI threshold and (ii) outperform their predicted scores given their economic complexity. Among high-income economies, global overperformers include Australia, Japan, the Netherlands, New Zealand, three of the four Asian Tigers (Hong Kong SAR, China; Singapore; and the Republic of Korea), and Northern European countries such as Denmark, Finland, and Norway—aligning with findings on the long-term determinants of AIPI detailed later. Upper-middle-income local overperformers comprise Albania, China, Costa Rica, Indonesia, Kazakhstan, Malaysia, and Ukraine. Finally, lower-middle- and low-income local overperformers include Ghana, India, Morocco, Rwanda, Sri Lanka, Tunisia, and Viet Nam.

From this list of AI overperformers, we observe that AI regulation and ethics consistently emerge as top drivers of AI preparedness across income groups, emphasizing the universal importance of institutions in managing AI integration. In low- and lower-middle-income countries (LICs/LMICs), regulation and workforce development dominate, reflecting foundational priorities amid resource constraints, while digital infrastructure and innovation receive less emphasis. For upper-middle-income countries (UMICs), regulation and human capital remain critical, but digital infrastructure becomes increasingly significant as technological capacities expand. High-income countries (HICs) prioritize regulation alongside well-established digital infrastructure and innovation systems, supported by advanced workforce readiness strategies.

This progression illustrates how AI preparedness strategies align with the unique capacities and priorities of each income group, offering tailored policy pathways for countries aiming to enhance their AI readiness. By understanding these patterns, policy makers can adopt fitted strategies to develop their countries' AI ecosystems based on their specific economic and institutional contexts and might effectively bridge the gap in the global AI divide.

The rest of the paper is organized as follows: Section II outlines the data and methodology, Section III presents the main empirical findings, Section IV examines how different categories of countries overperform in AI preparedness, and Section V concludes and opens avenue for future research on AI preparedness.

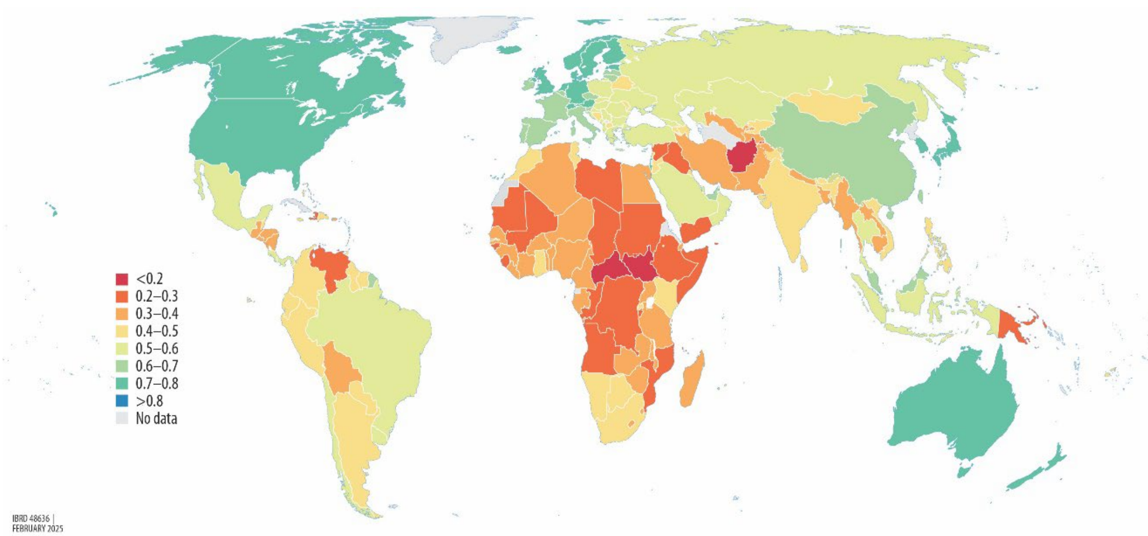
II- Data and Methodology

II.a- Data

The AIPI serves as our main output of interest and is designed by the IMF to assess the readiness of 174 countries to adopt and integrate AI technologies as of 2023 (Cazzaniga et al., 2024). It is built around four core pillars that are deemed essential for smooth AI adoption: (i) digital infrastructure, (ii) human capital and labor market policies, (iii) innovation and economic integration, and (iv) regulation and ethics. Each dimension is calculated by averaging a set of normalized sub-indicators that capture specific aspects relevant to AI readiness, such as internet accessibility (pillar i), STEM education (pillar ii), R&D investment (pillar iii), and adaptable legal frameworks (pillar iv).[6] The global AIPI index, scaled from 0 to 1, combines these four pillars, each contributing up to 0.25. Higher scores indicate greater AI preparedness, providing stakeholders with a benchmark for identifying areas needing improvement to enhance AI integration across economies. The AIPI focuses on AI adoption preparedness rather than invention leadership—dominated by China and the United States,[7] as highlighted in the "Draghi Report" —thereby enabling a comparative assessment of preparedness levels across all economies.

The AIPI distribution (Figure 1) reveals substantial disparities in AI preparedness and readiness across nations. Research underscores that AI readiness is influenced by a country's economic structure, infrastructure, and policy environment. For example, Liu and Wang (2024) find strong correlations between income level, digital infrastructure, and human capital with AI uptake, with higher-income nations demonstrating more advanced AI ecosystems.[8] These findings align with growing evidence of a widening divide between countries equipped to leverage AI and those at risk of being left behind (Cazzaniga et al., 2024).

Figure 1: Global distribution of AIPI (2023)



[6] More details here: https://www.imf.org/external/datamapper/AIPINote.pdf.

[7] European Commission (2024, Fig. 2 p. 36).

[8] Additionally, Liu and Wang (2024) show that countries with high English proficiency and a specialization in digitally tradable services have shown higher adoption rates, underscoring the importance of digital connectivity and linguistic compatibility with generative AI tools.

Source: IMF, AIPI data mapper [https://www.imf.org/external/datamapper/AI_PI@AIPI/ADVEC/EME/LIC]. Notes: The map is checked and validated by the World Bank Cartography Unit as of February 02, 2025.

A methodological decision in this study is the choice of the recent AIPI from the IMF over the AI Readiness Index developed by the private UK-based consulting firm Oxford Insights.[9] This decision is primarily driven by the IMF's reputation as a globally recognized multilateral institution and supposed to be accountable, which ensures greater scrutiny and transparency in the development of its products. In contrast, the outputs of private entities like Oxford Insights, though valuable, do not operate within the same level of oversight or accountability. Furthermore, the AI Readiness Index is explicitly designed to assess the extent to which governments are prepared to deploy AI in public service delivery.[10] In contrast, the AIPI aims to focus on a more comprehensive evaluation of how countries can harness AI capabilities effectively. That said, the choice of index does not significantly impact the broader contribution of this study, as the proposed methodology remains adaptable and replicable.

Employing Bayesian Model Averaging (BMA), we analyze a comprehensive set of 67 historical, political, and economic indicators from growth determinants literature to identify robust predictors of national AIPI. The BMA framework,[11] as implemented here, enables us to isolate the most robust determinants of AI preparedness from a wide range of potential predictors, as defined and built by Sala-i-Martin, Doppelhofer, and Miller (2004).[12] Figure 2 illustrates the inclusion probabilities of various covariates in the top 5,000 models. The results highlight that initial conditions—such as life expectancy, higher education levels, and economic size in 1960—along with cultural and geographic influences, including Confucian heritage and an East Asian regional dummy, are positively correlated with AIPI. Conversely, the indicator political instability, captured by the frequency of revolutions and military coups, shows a negative association with AIPI.[13]

Figure 2: Model Inclusion in BMA on AIPI [hyper-g/n prior; random model prior]

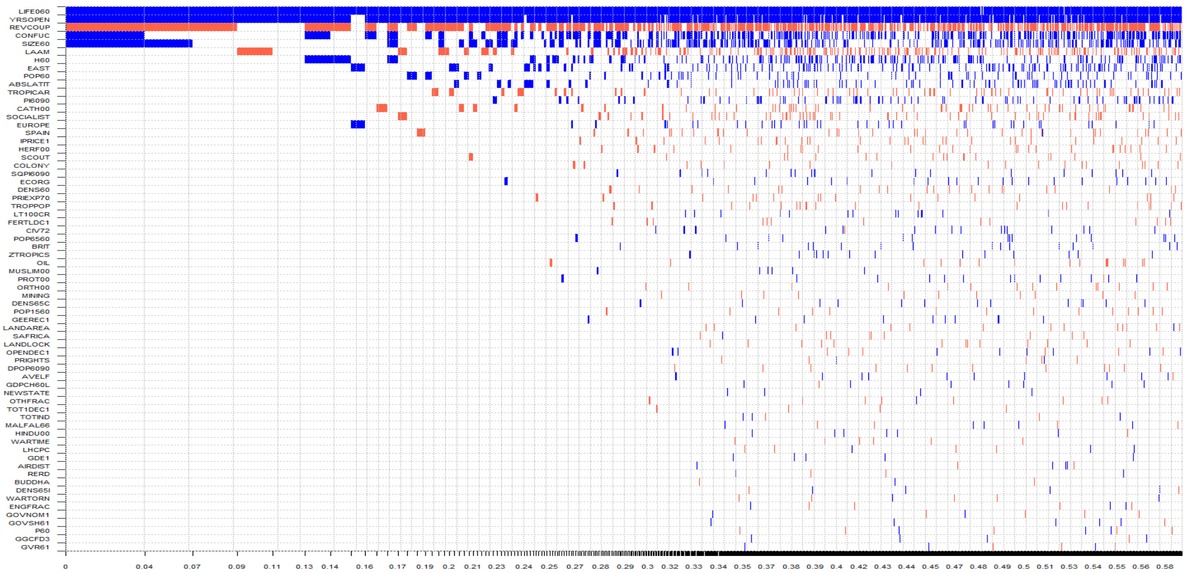
[9] See https://staging2.oxfordinsights.com/ai-readiness/.

[10] Ibid.

[11] BMA addresses the challenge of model uncertainty by allowing for the evaluation of multiple competing models rather than relying on a single model specification. In the context of this study, BMA provides a systematic way to assess the relative importance of various predictors of AI preparedness and readiness, through the AIPI, by calculating each variable's Posterior Inclusion Probability (PIP).

[12] Their database is available on the website of the American Economic Association (AEA): https://www.aeaweb.org/articles?id=10.1257/0002828042002570.

[13] Furthermore, historical trade openness emerges as a significant driver of AI adoption. However, this correlation should be interpreted cautiously, as one of the AIPI pillars (pillar iii: innovation and economic integration) directly captures aspects of trade openness and liberalization. The exact definitions of variables and their sources are available in (Sala-i-Martin, Doppelhofer, and Miller, 2004, p. 820-821).



Source: Author's construction, based on the AIPI and the database from Sala-i-Martin, Doppelhofer, and Miller (2004). The database is available here: https://www.openicpsr.org/openicpsr/project/116024/version/V1/view. Notes: The BMA considers a chain of three million recorded draws with one million burn-ins, by applying the birth-death sampler, using the BMS package from Zeugner and Feldkircher (2015). The figure depicts the results of BMA. The explanatory variables are ranked according to their Posterior Inclusion Probabilities (PIPs) from the highest on the top to the lowest at the bottom. The horizontal axis shows the values of cumulative posterior probability. Blue (darker in greyscale) and red (lighter in greyscale) colors denote the positive and negative sign of the estimated parameter of explanatory variable, respectively. No color means the corresponding explanatory variable is not included in the model.

The results from the BMA align with existing literature on the risks of a growing AI divide, emphasizing that established industrialized nations and East Asian economies, particularly those with Confucian cultural influences (i.e., China; Hong Kong SAR, China; Singapore; Korea; and Taiwan, China), consistently lead in AI preparedness and readiness. Conversely, politically unstable countries face significant challenges in this domain. This finding motivates the approach in the current paper to move beyond these general trends by identifying both global and local AI overperformers relative to the complexity of their economic structures. The goal is to uncover how countries can learn from these champions to effectively enhance their AI policy agendas.

To measure the economic complexity of each country, we consider ECIs for trade, technology, and research available on the Observatory of Economic Complexity (OEC) website.[14] However, since the ECI for technology, derived from patent data, is available for fewer countries (90 compared to 133 for ECI trade and 130 for ECI research), we perform a Principal Component Analysis (PCA) using only ECI trade and ECI research for 2021–22, the most recent years available. The first principal component, which explains over 50 percent of the variance, is used to represent the multidimensional economic complexity of each country. This component is then normalized and scaled between 0 and 1 across the sample of countries.

[14] See https://oec.world/en/rankings/eci/hs6/hs96?tab=ranking.

As highlighted by Hidalgo (2023), ECI offers valuable insights into a country's productive capabilities by analyzing the diversity and sophistication of its exports, and by extension, its academic research. The paper demonstrates how metrics derived from ECI, such as relatedness and complexity, can guide industrial policies and diversification strategies, positioning ECI as a critical tool for identifying opportunities for structural transformation and economic upgrading. ECI's relevance logically extends to capturing ability to adopt cutting-edge technology, such as AI preparedness and readiness, as it reflects a country's knowledge intensity, innovation capacity, and productive capabilities. Higher ECI scores suggest that economies are better equipped to adopt advanced technologies like AI due to their diversified industrial base, skilled workforce, and robust infrastructure.

Finally, the income levels of countries are classified using the World Bank's Atlas Method, incorporating updated income classifications for FY25 based on 2023 GNI per capita data.[15] This classification is crucial for distinguishing between global AI overperformers, likely concentrated among HICs, and local overperformers within LICs, LMICs, and UMICs.

II.b- Methodology

To identify AI global and local overperformers, we adopt a two-step approach: (i) identifying countries with AIPI scores above the weighted median and average scores of their respective income groups,[16] and (ii) determining those with AIPI scores exceeding their predicted values based on their economic complexity. Formally, a weighted least square approach is considered:

$$AIPI_I = \alpha + \beta ECI_i + \varepsilon_i, \quad (1)$$

where: $min_{\alpha,\beta} P_i(AIPI_i - (\alpha + \beta ECI_i))^{\,2}$ (2)

$$\widehat{AIPI_\iota} = \hat{\alpha} + \hat{\beta} ECI_i \quad (3)$$

The $\widehat{AIPI}$ index for country i is then predicted by an intercept $\hat{\alpha}$, and the first component of the ECI on trade and research $\hat{\beta}$, weighted by the population size of each country. Such weighting is crucial to ensure the real-world representativeness of AI preparedness and economic complexity index scores.[17] The ex-ante idiosyncratic error term (ε_i) or ex-post residual term ($\widehat{\varepsilon_\iota}$) captures respectively the theoretical and actual gap between observed $AIPI$ and predicted $\widehat{AIPI}$ scores. Accordingly, a country is classified as an overperformer if:

$$\begin{cases} AIPI_i > \widehat{AIPI_\iota}, or\ \widehat{\varepsilon_\iota} > 0 \\ AIPI_i > AIPI_{i,theshold} \end{cases} \quad (4)$$

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[15] See https://blogs.worldbank.org/en/opendata/world-bank-country-classifications-by-income-level-for-2024-2025.

[16] Average and median scores for income groups are calculated using population size for 2023 (“SP.POP.TOTL”), sourced from the World Bank's open data via the “wbopendata” module in Stata (Azevedo, 2020). Population data for Taiwan, China, is retrieved from the Statista website [https://www.statista.com/statistics/319793/taiwan-population/].

[17] Unweighted Ordinary Least Squares estimates do not alter the classification of overperforming countries and can be obtained from the replication material.

where $AIPI_{i,theshold}$ indicates the relevant average and median AIPI score for country i for relevant income categories.

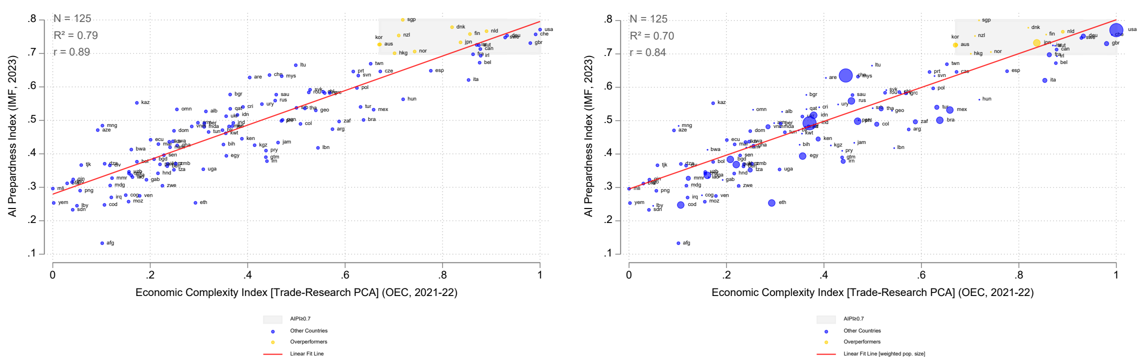
III- Empirical findings

III.a- Global overperformers

The analysis of the two charts below, one unweighted and the other weighted by population size, reveals important insights about the relationship between the AIPI and the ECI trade, research composite obtained from 125 matching countries at a global level (Figure 3).[18] A strong relationship emerges between AIPI and ECI, with R^2 values of 0.79 (unweighted) and 0.70 (weighted), indicating that approximately three-quarters of the variance in AIPI is explained by the variance in ECI, and vice versa. The partial correlations (r) of 0.89 (unweighted) and 0.84 (weighted) further indicate a very high positive linear correlation, and it is statically significant at 0.1 percent. These results highlight the strong (and expected) link between economic complexity and AI preparedness, demonstrating that greater economic sophistication—reflected in the diversity and complexity of exports and research output—significantly enhances a country's capacity to adopt and integrate AI technologies effectively, irrespective of population weighting.

Figure 3: Global relationship between AIPI and ECI

a) Unweighted analysis b) Population-weighted analysis



Source: The AIPI score for 2023 is obtained from the IMF website [https://www.imf.org/external/datamapper/datasets/AIPI] and the ECI on trade and research for 2021-22 are obtained from the OEC website [https://oec.world/en/rankings/eci/hs6/hs96?tab=ranking]. Notes: The first principal component (PCA) of ECI trade and ECI research, with an eigenvalue exceeding 50 percent, is used to capture the multidimensional aspects of national economic complexity. Population size for 2023, used for the weighted analysis, is sourced from the "wbopendata" module in Stata (Azevedo, 2020), with data for Taiwan Province of China obtained from the Statista website. All countries with an observed AIPI score exceeding 0.7 (the weighted average and median score for HICs) are highlighted in the grey quadrant. Among these, only countries with AIPI scores surpassing their predicted values (above the linear fit line) are marked in yellow and identified as global overperformers.

Both charts highlight the top performers in AI preparedness, concentrated in the northeast quadrants. Consistent with previous BMA findings and the critical role of initial conditions in AI readiness, these top performers are exclusively HICs. To identify global overperformers within this

[18] All scatter plots benefit from the "white_cividis" scheme from Naqvi (2022).

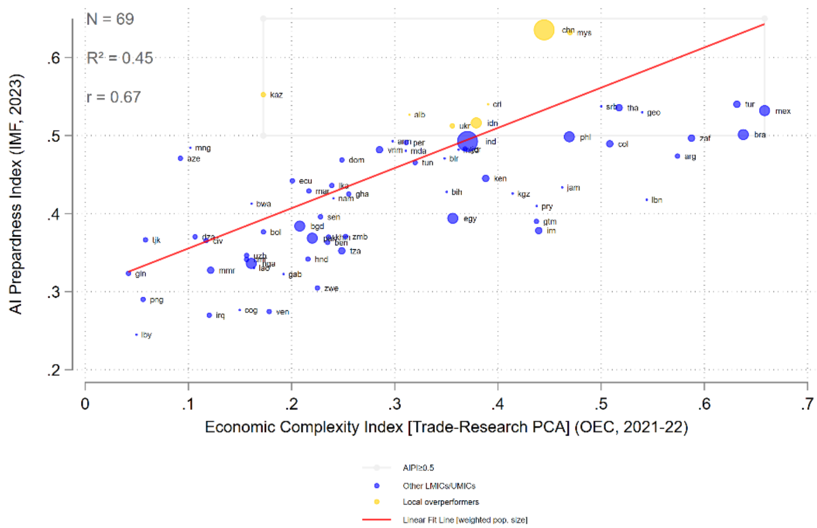
group, we focus on countries with an observed AIPI score exceeding 0.7—corresponding to the population-weighted average and median AIPI scores for HICs (rounded to one decimal)—and surpassing their predicted AIPI score (i.e., above the linear fit line) given their level of trade and research economic complexity. These 10 overperformers, marked in yellow, include Australia, Japan, the Netherlands, New Zealand, three of the four Asian Tigers (Hong Kong SAR, China; Singapore; and Korea), as well as Northern European countries such as Denmark, Finland, and Norway.[19] Their observed AIPI scores are statistically significantly above their predicted scores (at 0.1 percent threshold). The relative prominence of East Asian economies, particularly the three recently developed Asian Tigers and particularly Singapore—ranked #1 in observed AIPI score for 2023, much above its predicted score—aligns with prior BMA findings, which underscore the significant influence of "Confucian cultural values" on AI preparedness. By contrast, other HICs with strong AIPI scores, such as the United States and Switzerland, do not qualify as overperformers. This is because their observed AIPI scores align with the predictions based on their high levels of economic complexity rather than exceeding them. Additionally, the largest economies, like the United States, may face challenges in scaling top-level AI implementation uniformly across regions (Hunt, Cockburn, and Bessen, 2024).

III.b- Local overperformers

Figure 4, weighted by population size, demonstrates a positive but more moderate (yet still economically significant) correlation between AIPI and multidimensional economic complexity for the subgroup of 69 LMICs and UMICs, with an R^2 value of 0.45 and r of 0.67 (statically significant at 0.1 percent). This suggests that while economic complexity likely plays a role in shaping AI preparedness for these countries, policy decisions appear to have a more substantial independent impact on AIPI performance. A likely explanation, explored further in the next section, is that some of these countries, especially among UMICs, have greater macro-fiscal capacity than LICs (Choudhary, Ruch, and Skrok, 2024), enabling them to implement ambitious digital and AI-specific policy agendas. However, these countries often lack a fully developed infrastructure, and specifically digital infrastructure (pillar i of AIPI) and a sufficiently skilled workforce (pillar ii of AIPI), which are still in the process of being built, especially in comparison to HICs (World Bank, 2021; 2024c).

[19] With an impressive AIPI score of 0.76 in 2023 and an ambitious digital agenda, Estonia has the potential to be considered among the AI preparedness overperformers. However, a lack of data on its multidimensional economic complexity limits such classification. Estonia's third national AI Strategy (2024–26) prioritizes advancing AI skills and competencies through education and training reforms, including the introduction of new Master's and PhD programs, elective AI courses, and vocational education. The strategy also emphasizes society-wide upskilling initiatives, such as a digital academy and public sector training programs that aim to train 10 percent of employees annually. Additionally, the Ministry of Education and Research offers specialized AI training for teachers, students, and public sector managers, while the Ministry of Economic Affairs and Communications targets 80 percent AI literacy across society and widespread AI adoption—100 percent in the public sector and 75 percent in the private sector—by 2030 (Digital FOSS, 2024).

Figure 4: AI preparedness and economic complexity in LMICs and UMICs: Identifying local overperformers



Source: The AIPI score for 2023 is obtained from the IMF website [https://www.imf.org/external/datamapper/datasets/AIPI] and the ECI on trade and research for 2021-22 are obtained from the OEC website [https://oec.world/en/rankings/eci/hs6/hs96?tab=ranking]. Notes: The first principal component (PCA) of ECI trade and ECI research, with an eigenvalue exceeding 50 percent, is used to capture the multidimensional aspects of national economic complexity. Population size for 2023, used for the weighted analysis, is sourced from the "wbopendata" module in Stata (Azevedo, 2020). All countries with an observed AIPI score exceeding 0.5 (the weighted average and median score for the subgroup) are highlighted in the grey quadrant. Among these, only countries with AIPI scores surpassing their predicted values (above the linear fit line) are marked in yellow and identified as local overperformers.

Local overperformers are the seven countries with an observed AIPI score exceeding 0.5—equivalent to the population-weighted average and median AIPI scores for the subgroup (rounded to one decimal)—and surpassing their predicted AIPI scores based on their trade and research economic complexity. These overperformers include Albania, China, Costa Rica, Indonesia, Kazakhstan, Malaysia, and Ukraine. Their observed AIPI scores are statistically significantly above their predicted scores (at the 1 percent threshold). Among them, China and Malaysia stand out with the highest observed AIPI scores, reaffirming the strong AI readiness of East Asian countries in addition to industrialized nations. Kazakhstan also exhibits a notably high observed AIPI score relative to its economic complexity—its export basket being heavily reliant on hydrocarbons and mineral products, which limits its ECI trade score (ranked #76 out of 133 countries) and its research economic complexity is low (ranked #121 out of 130 countries) for the 2021–22 period. While the drivers of this overperformance will be examined further in the next section, it is important to emphasize China's critical role in the global AI landscape. As a co-leader in AI innovation alongside the United States (European Commission, 2024), China has made significant investments in chip supply chains,[20] and in digital infrastructure (de Seta, 2023)[21] and, in January 2024, introduced draft guidelines aimed at

[20] See https://www.economist.com/business/2024/01/01/welcome-to-the-era-of-ai-nationalism.
[21] For a deep dive on the infrastructure and superstructure of the "social credit" system (社会信用体系 shehui xinyong tixi), see Durand (2024, chap. 2).

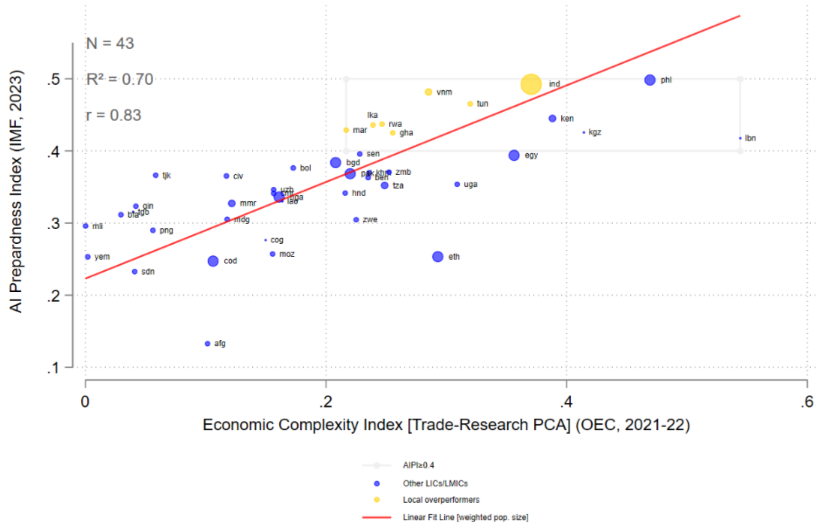
standardizing the AI sector (PRC MIIT, 2024).[22] Furthermore, in December 2024, China's Ministry of Industry and Information Technology (MIIT) established an artificial intelligence standardization technical committee. This committee is tasked with developing industry standards in areas such as large language models (LLMs) and AI risk assessment. Lastly, DeepSeek-R1, an open-source reasoning model developed by the Chinese AI startup DeepSeek (or 深度求索) was publicly released on January 20, 2025. A notable aspect of this LLM is its cost-effectiveness. The model was developed with an investment of under US$6 million, a fraction of the expenditure—estimated to be multiple billions—reportedly associated with training models like OpenAI's o1. It also relies on inferior chips.[23] By contrast, certain UMICs with relatively high observed AIPI scores, such as Brazil, Mexico and Türkiye, appear to underperform relative to their multidimensional ECI.

Last but not least, Figure 5, also weighted by population size, highlights the relationship between AIPI and multidimensional economic complexity for the subgroup of 43 LICs and LMICs, with an R^2 value of 0.70 and r of 0.83 (statically significant at 0.1 percent). This stronger correlation aligns with the earlier explanation regarding fiscal buffers. Unlike certain UMICs, these countries, particularly LICs, face severe fiscal constraints, including notably low levels of tax mobilization, which significantly limit their policy options (Choudhary, Ruch, and Skrok, 2024). As a result, they are compelled to prioritize cost-effective digital and AI-specific strategies whenever feasible. This also highlights that, similar to HICs at the opposite end of the spectrum, their AI preparedness is on average very closely linked to their trade and research complexity. However, this relationship functions at a much lower equilibrium, constrained by their critical multidimensional development challenges (UNSD, 2024).

[22] These guidelines propose the development of over 50 national and industry-wide AI standards by 2026 and contributions to more than 20 international AI standards, focusing on key technologies and practical applications. This comprehensive approach underscores China's strategic commitment to AI dominance.

[23] "In terms of performance, DeepSeek-R1 has demonstrated capabilities comparable to leading models like OpenAI's o1. For instance, on the MATH-500 benchmark, which assesses high-school-level mathematical problem-solving, DeepSeek-R1 achieved a 97.3 percent accuracy rate, slightly outperforming OpenAI o1's 96.4 percent. In coding tasks, it scored 49.2 percent on the SWE-bench Verified benchmark, edging out OpenAI o1's 48.9 percent." [https://venturebeat.com/ai/calm-down-deepseek-r1-is-great-but-chatgpts-product-advantage-is-far-from-over].

Figure 5: AI preparedness and economic complexity in LICs and LMICs: Identifying local overperformers



Source: The AIPI score for 2023 is obtained from the IMF website [https://www.imf.org/external/datamapper/datasets/AIPI] and the ECI on trade and research for 2021-22 are obtained from the OEC website [https://oec.world/en/rankings/eci/hs6/hs96?tab=ranking]. Notes: The first principal component (PCA) of ECI trade and ECI research, with an eigenvalue exceeding 50 percent, is used to capture the multidimensional aspects of national economic complexity. Population size for 2023, used for the weighted analysis, is sourced from the "wbopendata" module in Stata (Azevedo, 2020). All countries with an observed AIPI score exceeding 0.4 (the weighted average and median score for the subgroup) are highlighted in the grey quadrant. Among these, only countries with AIPI scores surpassing their predicted values (above the linear fit line) are marked in yellow and identified as local overperformers.

Nonetheless, seven countries have been identified as overperformers in AI preparedness, with an observed AIPI score exceeding 0.4—equivalent to the population-weighted average and median AIPI scores for the subgroup (rounded to one decimal): Ghana, India, Morocco, Tunisia, Rwanda, Sri Lanka, and Viet Nam. Their observed AIPI scores are statistically significantly above their predicted scores (at the 0.1 percent threshold). While the inclusion of countries like India, Tunisia, and Viet Nam is consistent with their well-established digital agendas, which emphasize leveraging technology for economic growth, institutional improvements, and social development, Rwanda's presence is particularly striking as the only LIC among the overperformers. Rwanda's strong performance can be partly attributed to its high domestic revenue mobilization, supported by effective donor engagement, and a vertically integrated decisional structure that facilitates bold reforms (Todd, 2019; IMF, 2023).[24] This fiscal capacity, coupled with a high Country Policy and Institutional Assessment (CPIA) score (World Bank, 2024b), so a substantial International Development Association (IDA) envelope allocation, and a business-friendly environment (World Bank 2024a), enables the country to advance an ambitious digital agenda. A notable example of this commitment is Rwanda's Ministry of Information Communication Technology and Innovation (MINICT) co-developing the AI Playbook for Small States in collaboration with Singapore's Ministry of Digital Development and Information under the Digital Forum of Small States initiative (Digital FOSS,

[24] Todd (2019) includes a few dedicated sections on Rwanda but further explores its traditional family system, specifically the stem family model and its implications. The relatively small size of the country (approximately 26,000 km^2) also likely contributes to more efficient revenue collection.

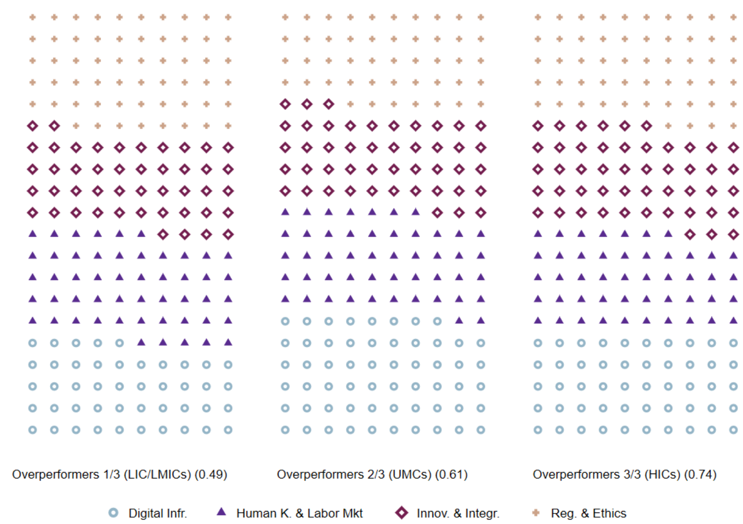
2024). In contrast, a few LMICs with relatively high observed AIPI scores, such as Kenya, Kyrgyzstan, Lebanon, and the Philippines, fall short of the performance levels suggested by their multidimensional ECI.

IV- Decoding overperformance: Insights into AI preparedness success

IV.a- Unpacking the Pillars of Overperformance across Income Groups

Figure 6 provides a comparative visual representation of the pillars driving AI preparedness across the three groups of overperformers. Each income group exhibits distinct patterns in the emphasis placed on the four pillars of AI preparedness described in Section II: digital infrastructure (pillar i), human capital and labor markets (pillar ii), innovation and integration (pillar iii), and regulation and ethics (pillar iv).

Figure 6: Key pillars driving overperformance across income groups of overperformers



	Overperformers		
	LIC/LMICs	UMICs	HICs
Digital Infr.	22.0%	28.8%	24.7%
Human K. & Labor Mkt	25.3%	24.0%	23.1%
Innov. & Integr.	22.6%	22.7%	24.3%
Reg. & Ethics	30.1%	24.5%	27.9%

Source: The AIPI score and pillars for 2023 are obtained from the IMF website [https://www.imf.org/external/datamapper/datasets/AIPI].
Notes: Population size for 2023, used for the weighted analysis, is sourced from the "wbopendata" module in Stata (Azevedo, 2020). Weighted average AIPI scores per group in parenthesis. Each symbol represents 0.5 percent of the total AIPI score per group. Digital Infrastructure (pillar i, ○): Investments and capabilities in foundational technologies to support AI readiness. Human Capital & Labor Markets (pillar ii, ▲): Workforce skills and readiness for AI-driven economies. Innovation & Integration (pillar iii, ◇): The ability to innovate and integrate AI into industries and economic activities. Regulation & Ethics (pillar iv, +): Institutional frameworks and ethical guidelines to support responsible AI deployment. The "waffle" package is taken from Naqvi (2024b).

In low-income and lower-middle-income overperformers, regulation and ethics emerge as the most significant driver of AI preparedness, accounting for 30.1 percent of the emphasis (Indian and Sub-

Saharan Africa/SSA models as detailed hereafter). This highlights the critical role of institutions in building trust and establishing frameworks for AI deployment, even amid considerable structural challenges. Following closely is the focus on human capital and labor markets (25.3 percent, Indian model), reflecting the pressing need to develop skills and prepare the workforce for a technology-driven future. However, digital infrastructure (22.0 percent) and innovation and integration (22.6 percent) are less emphasized, likely constrained by limited fiscal resources and technological investment capacity.

For upper-middle-income overperformers, the emphasis shifts, with digital infrastructure taking the lead (28.8 percent, Chinese model), reflecting the growing technological capabilities and infrastructure investments of these nations. Regulation and ethics remain essential at 24.5 percent (Malaysian model), underscoring the importance of robust institutional framework in fostering AI readiness. Human capital and labor markets (24.0 percent, Malaysian and Kazakh model) also maintain a strong focus as these nations work to enhance workforce preparedness. Innovation and integration, however, remain the least prioritized pillar at 22.7 percent, possibly due to structural constraints, though this requires careful interpretation. For instance, China's status may be underestimated in this metric due to the "economic integration" component, which accounts for tariff and non-tariff barriers, as well as restrictions on capital and labor movement. This underscores that robust domestic innovation can thrive even with lower scores in economic integration measures.

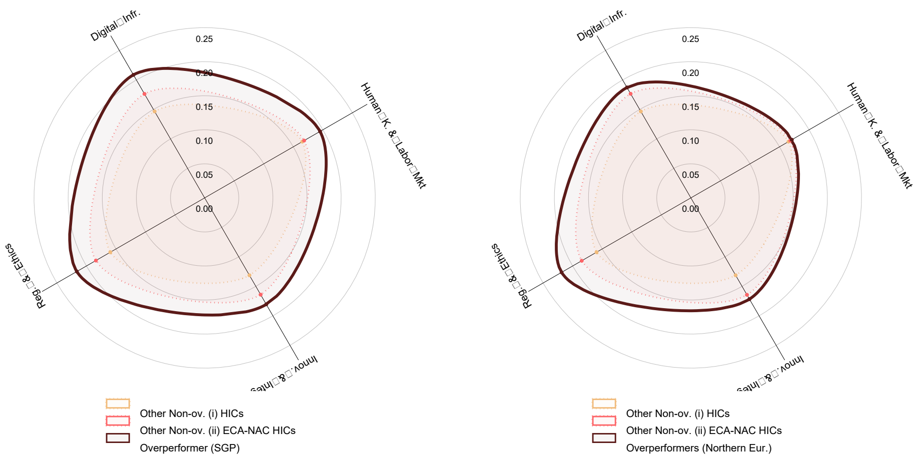
High-income overperformers present a more balanced profile, with regulation and ethics taking the lead at 27.9 percent (Singapore and North Europe model), emphasizing the importance of institutional frameworks in managing advanced AI integration. Digital infrastructure follows at 24.7% (Singapore model), reflecting the continuous effort to maintain a technological edge. Innovation and integration (24.3 percent) gain slightly more prominence compared to lower-income groups, while human capital and labor markets (23.1 percent) play a supporting role, as these countries already benefit from well-established systems for education and workforce development.

Across all income levels, regulation and ethics consistently top the list, but the relative emphasis on digital infrastructure, workforce development, and innovation adapts to the specific needs and capacities of each economic group.

IV.b- Selected country cases

Figure 7: Selected HICs (global) overperformers

a) Singapore b) Northern Europe (Denmark, Finland, Norway)



Source: The AIPI score and pillars for 2023 are obtained from the IMF website [https://www.imf.org/external/datamapper/datasets/AIPI]. Notes: Population size for 2023, used for the weighted analysis, is sourced from the "wbopendata" module in Stata (Azevedo, 2020). Weighted average AIPI scores per group in parenthesis. Dotted yellow shows the weighted average pillar scores for non-overperformer HICs in Europe, Central Asia, or North America, while dotted red shows the scores for all other non-overperformer HICs. The "spider" package is taken from Naqvi (2024a).

Singapore's AI readiness strategy strongly emphasizes all the AIPI pillars detailed above, and particularly regulation and ethics, as well as the development of robust digital infrastructure and solid human capital and labor market policies to enable responsible and efficient AI adoption (Figure 7a).[25]

Singapore's "Model AI Governance Framework," developed by the Infocomm Media Development Authority (IMDA) and the Personal Data Protection Commission (PDPC), is a cornerstone of its AI strategy.[26] This framework provides practical guidelines for deploying AI ethically, with a focus on principles such as transparency, fairness, and accountability. To ensure these principles are actionable, Singapore launched AI Verify, a world-first AI testing framework and toolkit that enables organizations to validate the trustworthiness of their AI systems against regulation standards. These efforts reflect Singapore's commitment to balancing innovation with public trust, ensuring AI systems are both ethical and effective.

Under its first National AI Strategy or NAIS (Smart Nation Singapore, 2019), about two years before the release of GPT-3.5 by OpenAI, Singapore emphasized heavy investments in digital infrastructure

[25] Singapore also excels in AI human capital and labor market development, particularly in scaling up AI talent across creators, practitioners, and users to meet the growing demands of AI adoption. Its latest national strategy (Smart Nation Singapore, 2023) prioritizes the cultivation of top-tier AI researchers and engineers through tailored support, redesigned training programs, and global talent attraction initiatives. Programs like the AI Apprenticeship Program and sector-specific upskilling efforts under the Industry Transformation Maps (ITMs) play a central role in reskilling the workforce for AI integration. By leveraging targeted interventions and fostering industry-academia collaborations, Singapore aims to enhance workforce readiness, ensuring that its citizens can thrive in an AI-driven economy.

[26] See https://www.pdpc.gov.sg/help-and-resources/2020/01/model-ai-governance-framework.

to create a foundation for widespread AI adoption. Key initiatives included expanding high-speed connectivity networks and building state-of-the-art data centers to support AI applications requiring significant computational and data-sharing capabilities. The government also supported the integration of AI across nine priority sectors such as health care, transport, and finance[27] by creating sector-specific data ecosystems and AI solutions. The orientations are reaffirmed in the NAIS 2.0 (Smart Nation Singapore, 2023). This infrastructure ensures Singapore can effectively deploy AI at scale while maintaining seamless interoperability and efficiency.

Northern Europe (i.e., Denmark, Finland, and Norway), has developed comprehensive AI readiness strategies that prioritize regulation and ethics to ensure responsible AI development and deployment (Figure 7b).

Denmark has established ethical principles for AI, as part of its national strategy to maintain high levels of trust in technology (Danish Gvt, 2019).[28] These principles reflect "Danish values" and are based on recommendations from an official expert group on data ethics, as well as draft EU ethical guidelines for AI. The framework is designed to guide both public authorities and businesses in the responsible use of AI, ensuring that ethical considerations are integrated into AI development and application.

The "Artificial Intelligence 4.0 Programme" highlights Finland's focus on robust regulation and ethics as central pillars in its AI strategy (MEAEE, 2022). The program emphasizes the importance of balancing digital and green transitions, underpinned by principles of sustainable development and technological sovereignty. By prioritizing ethics and regulation, Finland seeks to establish trust in AI technologies, ensure responsible deployment, and position itself as a frontrunner in creating fair and sustainable AI-driven solutions for global markets. The approach includes collaboration involving public, private, and research sectors to align digital innovation with societal and environmental goals.

The Norwegian "Digitalization Strategy 2024-2030" emphasizes a holistic approach to policy making and ethics in the context of AI readiness and digital transformation (MDPG, 2024). The regulatory framework includes ensuring robust privacy protection, secure digital infrastructure, and transparent use of AI technologies aligned with ethical standards. It also prioritizes fostering public trust through inclusive digital practices and safeguarding against digital vulnerabilities such as fraud and misinformation. The strategy integrates AI into broader societal and economic objectives, reinforcing oversight structures to create equitable opportunities while addressing societal challenges like security, environmental sustainability, and the ethical use of advanced technologies.

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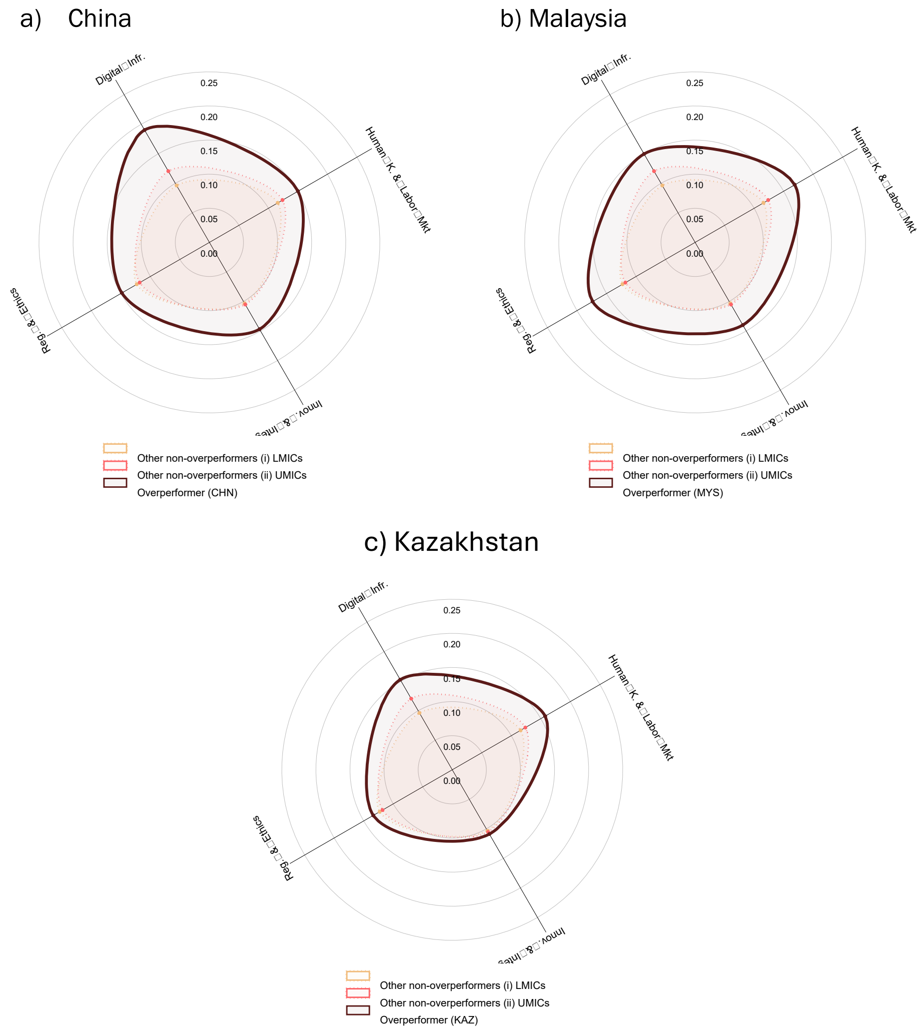
[27] "We have identified nine such sectors: Transport & Logistics, Manufacturing, Finance, Safety & Security, Cybersecurity, Smart Cities & Estates, Healthcare, Education and Government." (Smart Nation Singapore, 2019, p.16)

[28] There are six ethical principles to guide responsible AI development and use: (i) self-determination ensures citizens can make informed, independent choices when interacting with AI; (ii) human dignity emphasizes respecting human rights and diversity; (iii) equality and justice ensure fairness and non-discrimination in AI systems; (iv) responsibility promotes accountability in AI development and deployment; (v) explainability advocates for transparency to build trust by clarifying how AI decisions are made; and (vi) ethically responsible development mandates that AI aligns with societal values and ethical standards. These principles aim to collectively foster trust and ensure that AI benefits society.

Figure 8: Selected UMICs overperformers



Source: The AIPI score and pillars for 2023 are obtained from the IMF website [https://www.imf.org/external/datamapper/datasets/AIPI].
Notes: Population size for 2023, used for the weighted analysis, is sourced from the "wbopendata" module in Stata (Azevedo, 2020). Weighted average AIPI scores per group in parenthesis. Dotted yellow shows the weighted average pillar scores for non-overperforming LMICs and dotted red shows them for non-overperforming UMICs. The "spider" package is taken from Naqvi (2024a).

Spiderweb charts illustrate that China and Malaysia excel in all AIPI pillars relative to the average of non-overperformers within their income group. Notably, China demonstrates exceptional proficiency in digital infrastructure (Figure 8a), whereas Malaysia shows significant strengths in regulation and ethics, as well as human capital and labor market policies (Figure 8b).

As already underlined, China's AI readiness strategy heavily emphasizes the development of robust digital infrastructure as a cornerstone for its technological ambitions (de Seta, 2023). Central to this effort is the "New Generation Artificial Intelligence Development Plan" or AIDP (China State Council, 2017), which outlines a roadmap to position China as a global AI leader by 2030, underscoring the critical role of advanced infrastructure.[29] Through significant investments, China has built the world's largest 5G network and an extensive fiber-optic broadband system, enabling high-speed connectivity and the Internet of Things (IoT) essential for AI applications across industries. Complementing this, initiatives such as the Digital Silk Road under the Belt and Road Initiative aim to enhance global digital connectivity, extending China's AI influence internationally. To meet its domestic AI computing needs, China's MIIT is advancing plans to bolster computational power and develop indigenous AI technologies by 2027.[30] Furthermore, China's smart city initiatives, exemplified by projects like City Brain, leverage AI and advanced infrastructure to optimize urban management and services (Zhang et al., 2019). While these efforts have positioned China as a leader in digital infrastructure, challenges such as regional disparities and regulatory complexities remain, prompting continuous efforts to ensure balanced and sustainable development,[31] as also observed in the United States, for instance (Hunt, Cockburn, and Bessen, 2024).

Malaysia has implemented robust frameworks to ensure responsible AI adoption, aligning with ethical principles. Its "AI Roadmap," or AI-Rmap (MOSTI, 2021) establishes guidelines to safeguard data privacy, promote transparency, and mitigate risks such as cybersecurity threats. Institutional frameworks like the AI-Catalyst coordinate implementation across government, industry, and academia. These efforts are reinforced by the "National Fourth Industrial (4IR) Policy" (EPU, 2021), which integrates anticipatory and agile regulatory approaches, ensuring that legislation adapts to evolving technological needs.

In September 2024, the country introduced its "National Guidelines on AI Governance & Ethics," or AIGE (MOSTI, 2024). These guidelines provide a comprehensive framework for stakeholders—including end users, policy makers, and developers—to adopt AI technologies in a safe, transparent, and ethical manner. The AIGE outlines seven fundamental principles designed to promote responsible AI practices across various sectors (e.g., finance, health care, manufacturing).[32]

To further strengthen its AI framework, Malaysia launched the National AI Office (NAIO) in December 2024. The NAIO serves as a centralized agency dedicated to formulating AI policies, addressing regulatory issues, and coordinating strategic planning and research.

Malaysia also emphasizes the development of a skilled workforce to drive AI adoption. The AI-Rmap (MOSTI, 2021) identifies fostering AI talents as a core strategy, focusing on reskilling and upskilling programs and partnerships with universities to cultivate industry-ready talent. Additionally, the National 4IR Policy (EPU, 2021) supports initiatives to match talent pipelines with future economic needs and reduce disparities in access to 4IR opportunities.

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[29] An unofficial translation of the AIDP is available on Stanford's DigiChina website: https://digichina.stanford.edu/work/full-translation-chinas-new-generation-artificial-intelligence-development-plan-2017/.

[30] See https://www.theregister.com/2024/01/31/china_homegrown_ai_infrastucture.

[31] See https://www.reuters.com/markets/asia/china-tackle-risks-new-infrastructure-push-2024-09-04.

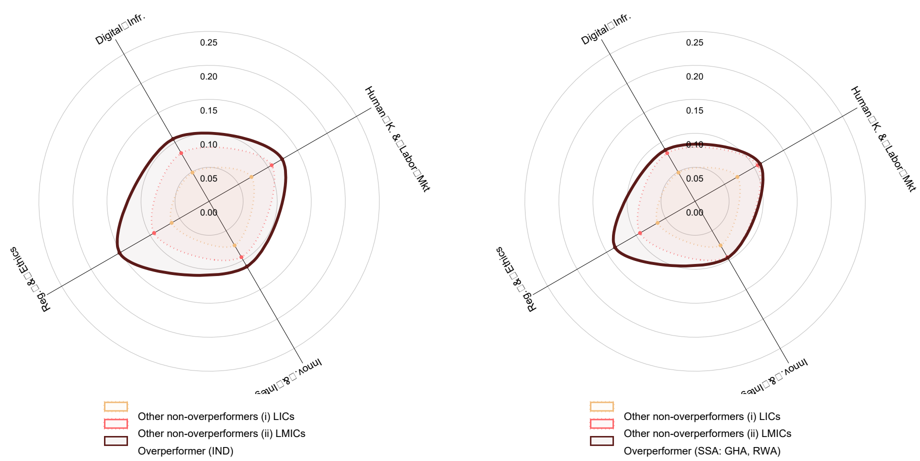
[32] These include (i) fairness, to prevent bias and ensure equitable access; (ii) reliability and safety, to guarantee secure and robust systems; (iii) privacy and security, to protect personal data and ensure informed consent; (iv) inclusiveness, to promote accessibility and diversity; (v) transparency, to make AI decision-making clear and understandable; (vi) accountability, ensuring developers and operators take responsibility for AI outcomes; and (vii) the pursuit of human benefit, prioritizing societal progress and well-being.

The Human Resources Ministry (KESUMA) recently conducted a comprehensive study on the impact of AI, digitalization, and the green economy across 10 strategic sectors.[33] The findings, released in September 2024, identified over 600,000 jobs requiring skill enhancements and highlighted 60 emerging roles, such as AI engineers and sustainability specialists, essential for advancing Malaysia's digital and green economy.[34]

Finally, Kazakhstan's "National Artificial Intelligence Development Concept for 2024–2029" (MDDIAI, 2024)[35] outlines a comprehensive strategy to position the country as a regional leader in AI, with a strong emphasis on human capital and labor market policies (Figure 8c). The strategy prioritizes human capital development through AI-focused education, reskilling programs, and widespread AI literacy campaigns, aiming to train 1 million citizens in AI skills. It emphasizes fostering research and innovation, including the development of a LLM to preserve Kazakh cultural and linguistic heritage, alongside investments in robust digital infrastructure. By integrating AI into traditional industries like agriculture and mining and promoting economic modernization, Kazakhstan aims to enhance productivity and global competitiveness while addressing labor market adaptation to technological advancements.

Figure 9: Selected LICs/LMICs overperformers

a) India | b) SSA (Ghana, Rwanda)



Source: The AIPI score and pillars for 2023 are obtained from the IMF website [https://www.imf.org/external/datamapper/datasets/AIPI].
Notes: Population size for 2023, used for the weighted analysis, is sourced from the "wbopendata" module in Stata (Azevedo, 2020). Weighted average AIPI scores per group in parenthesis. Dotted yellow shows the weighted average pillar scores for non-overperforming LICs and dotted red shows them for non-overperforming LMICs. The "spider" package is taken from Naqvi (2024a).

[33] See https://www.mida.gov.my/mida-news/ai-digitalisation-and-green-economy-to-reshape-malaysian-job-market-sim. "The report highlighted 10 key sectors in the first phase of the upskilling initiative – aerospace, chemicals, electrical and electronics, energy and power, food manufacturing and services, global business services, information and communications technology, medical devices, pharmaceutical manufacturing, and wholesale and retail trade.

[34] To support workforce upskilling and reskilling, KESUMA has allocated a RM3 billion fund (approximately US$670 million) to provide support through levies, credits, scholarships, and matching grants. These initiatives will be facilitated via the MyMAHIR platform and the forthcoming Akademi KESUMA. See https://hr.asia/asean/malaysia-to-study-impact-of-ai-digitalisation-and-green-economy-on-workforce.

[35] The Concept is available online and has been translated into English using Google's automatic translation tool: https://adilet.zan.kz/rus/docs/P2400000592.

India’s AI readiness strategy places a significant emphasis on both human capital development and an ethical framework to support its vision of becoming a leader in AI innovation (Figure 9a). On the human capital and labor market front, the IndiaAI Expert Group Report (MeitY, 2023) outlines a comprehensive approach that includes establishing Centers of Excellence (CoEs) for AI research and education, fostering industry-academic partnerships, and promoting entrepreneurship. The report emphasizes large-scale upskilling initiatives, such as the FutureSkills Program, which aims to build a robust AI talent pool through specialized curricula, faculty training, and career pathways for roles like data scientists and AI ethicists.[36] These interventions target not only IT professionals but also school students, graduates, and non-IT sectors, ensuring widespread AI literacy. Similarly, the National Strategy for Artificial Intelligence (NITI Aayog, 2018) highlights the importance of democratizing AI education through public-private partnerships and certification programs with market value, aiming to position India as an “AI Garage" capable of serving global AI needs.[37]

In terms of regulation and ethics, both documents prioritize the establishment of a robust framework to ensure responsible AI deployment. The NITI Aayog Strategy (NITI Aayog, 2018) advocates for “Responsible AI for All," emphasizing fairness, accountability, and transparency (FAT) as guiding principles. It proposes the creation of Ethics Councils within Centers of Research Excellence (COREs) to monitor compliance with ethical standards.[38] Data privacy and protection frameworks aligned with international norms are highlighted to mitigate risks related to bias, privacy violations, and security breaches. The IndiaAI Expert Group Report (MeitY, 2023) reinforces this by proposing a secure and transparent India Dataset Platform (IDP) to support ethical data usage while ensuring public trust through anonymization and secure protocols.[39]

Ghana and Rwanda are two SSA countries with notable performance in the regulation and ethics aspect of their AI preparedness strategies, indicating their efforts to develop responsible and inclusive AI ecosystems (Figure 9b). In Ghana’s National Artificial Intelligence Strategy (MOCD, 2022), the establishment of a Responsible AI Office (RAI Office) is a key component to coordinate stakeholders, monitor policy implementation, and promote trustworthy AI development.[40] The strategy emphasizes data privacy, transparency, and the creation of a regulatory framework to mitigate algorithmic bias and ensure secure data sharing. Ghana’s approach includes enforcing a robust data framework through initiatives like the Ghana Open Data Initiative (GODI), aimed at

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[36] The FutureSkills Program, launched by Indian National Association of Software and Service Companies (NASSCOM) in partnership with the Ministry of Electronics and Information Technology (MeitY), aims to build digital competencies in emerging technologies such as AI, data analytics, and cybersecurity. It offers industry-recognized certifications, virtual labs, and government incentives to make skilling affordable, with over 1.8 million learners benefiting from its flexible online and classroom formats.

[37] The AI Garage concept aims to position India as a global hub for AI innovation, where solutions developed for domestic challenges—such as health care, agriculture, and education—can be scaled and adapted for global markets, particularly for developing economies with similar needs.

[38] COREs are envisioned as specialized hubs for AI research, fostering innovation through collaborations between academia, industry, and government. These centers aim to advance cutting-edge AI solutions, promote ethical AI development through dedicated ethics councils, and build a robust talent pipeline to support India's AI ecosystem.

[39] The IDP is envisioned as a centralized, secure repository for high-quality datasets to support AI research and innovation. The platform aims to ensure responsible data sharing by incorporating robust privacy protections, data anonymization protocols, and frameworks that foster public trust and facilitate collaboration across academia, industry, and government.

[40] The RAI Office is a coordinating body aiming to oversee the implementation of AI policies, ensure compliance with ethical guidelines, and foster collaboration between government, academia, and industry. Its role includes monitoring AI applications to prevent bias and promoting transparency and public trust in AI systems.

supporting responsible AI adoption while safeguarding human rights.[41] Similarly, Rwanda's National AI Policy (MINICT, 2023) underscores practical ethical guidelines to align AI solutions with human rights and societal values. The policy mandates an annual review of these guidelines by the Rwanda Utilities Regulatory Authority (RURA) and incorporates fairness, transparency, and accountability in public sector applications.[42] Rwanda also integrates AI ethics into national education and collaborates with global institutions to align with international standards. The establishment of a Responsible AI Office (RAIO) further reinforces oversight and promotes ethical practices.[43] Both countries' strategies highlight a commitment to using frameworks to balance technological progress with social accountability, ensuring that AI advancements contribute to inclusive and sustainable development.

V- Discussion

The findings presented here underscore the multifaceted nature of AI preparedness and highlight the importance of tailored strategies that reflect the unique socio-economic and institutional realities of individual nations. The proposed methodology, which combines the AIPI with multidimensional ECI, offers a nuanced framework for assessing national AI readiness relative to economic complexity capacities. By identifying both global and local overperformers, this approach not only captures excellence at the highest level, but also elevates the achievements of countries that have outperformed their predicted potential despite economic constraints.

A recurring theme in global policy discussions on AI is the widening readiness gap between high-income nations and lower-income economies (Cazzaniga et al., 2024). While this AI divide highlights disparities in digital infrastructure, access to data, and human capital, it often overlooks the strides made by local "champions"—countries that, despite resource limitations, have implemented forward-thinking strategies to bolster their AI ecosystems. By shifting the focus from the divide itself to identifying and following the examples set by these overperformers, the proposed methodology reframes the narrative. It encourages policy makers to learn from the experiences of peer nations that have successfully leveraged regulatory coherence, ethical frameworks, and targeted investments to enhance their AI capacities.

For example, Rwanda and Ghana seem to demonstrate that robust regulation and ethics structures, combined with strategic partnerships and capacity-building initiatives, can compensate for limited economic complexity and fiscal constraints. Similarly, Kazakhstan's seemingly ambitious AI literacy and upskilling programs illustrate how countries can focus on workforce transformation to align their economic potential with AI advancements. These examples emphasize the importance of fostering

[41] The GODI is a national program aimed at enhancing data transparency and accessibility by making government datasets open to the public. It supports responsible AI adoption by providing high-quality data for research and innovation, while also ensuring data protection measures are in place to uphold privacy rights and prevent misuse.

[42] RURA is an independent multi-sector regulatory body responsible for overseeing utilities, including ICT and AI framework. RURA is tasked with conducting annual reviews of AI ethical guidelines, ensuring compliance with national policies, and addressing concerns related to fairness, accountability, and transparency in AI deployments.

[43] Rwanda's RAIO is a dedicated office established to promote responsible AI practices and foster ethical innovation. The RAIO aims to coordinate policy implementation, support regulatory reforms, and monitor AI systems for alignment with national guidelines, ensuring that AI applications foster inclusive development while adhering to global best practices.

innovation ecosystems that support sustainable AI adoption rather than merely seeking to close the divide.

The identification of both global and local overperformers encourages a peer-learning model where countries facing similar structural challenges can exchange best practices and tailor their AI readiness strategies. By promoting knowledge-sharing and collaboration, the framework proposed here helps transcend income-based categorizations and instead fosters a more comprehensive AI development landscape. This peer-learning approach also supports policy experimentation, enabling nations to test scalable, cost-effective solutions before committing to large-scale deployments. By spotlighting overperformers across income groups, the methodology provides actionable insights into the pillars—whether regulatory, educational, or infrastructural—that drive AI overperformance.

In this study, the methodological decision to use the recent AIPI from the IMF instead of the AI Readiness Index developed by the private UK-based consulting firm Oxford Insights is based on several considerations. The IMF, as a globally recognized multilateral institution, offers a higher level of accountability, scrutiny, and transparency in the development of its products. Conversely, while the contributions of private entities like Oxford Insights are valuable, they do not operate under the same oversight or accountability. Moreover, the AI Readiness Index specifically assesses the preparedness of governments to deploy AI in public service delivery, whereas the AIPI provides a more comprehensive evaluation of how countries can effectively harness AI capabilities. That said, the choice of index does not significantly impact the broader contribution of this study, as the proposed methodology remains adaptable and replicable. The focus is on providing a transparent, data-driven approach that allows policy makers to assess their country's AI preparedness relative to their economic complexity. By prioritizing a clear and simple framework, this study enables comparisons across different datasets and indices, reinforcing the robustness and utility of its findings regardless of the specific AI readiness index used.

Future research could also explore ways to improve the AIPI or similar indexes. For instance, our findings suggest that a significant number of East Asian countries—particularly those influenced by Confucian traditions—tend to overperform in AI preparedness regardless of their income levels, from Viet Nam to Singapore and China. This trend warrants further investigation to determine whether it reflects specific cultural norms and values that prioritize "social harmony" and collective design (Kung and Ma, 2014) that are appropriate for AI readiness, and/or if it is influenced by higher average cognitive abilities in these countries (Lynn and Meisenberg, 2010; Jones, 2011). Notably, collective cognitive abilities have been shown to positively impact economic performance (Madsen, 2016) and could be better captured in future iterations of the AIPI if they prove relevant for readiness in cutting-edge technologies like AI.

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Appendix:

Table A.1: Macro covariates and AI preparedness [BMA]

	Post Mean	Post SD	Cond.Pos.Sign	PIP
(Intercept)	-0.009	NA	NA	1.000
LIFE060	0.007	0.002	1.000	0.998
YRSOPEN	0.145	0.049	1.000	0.946
REVCOUP	-0.091	0.068	0.000	0.690
CONFUC	0.119	0.157	1.000	0.403
SIZE60	0.006	0.009	1.000	0.360
LAAM	-0.013	0.027	0.000	0.200
H60	0.117	0.263	1.000	0.188
EAST	0.012	0.035	1.000	0.122
POP60	0.000	0.000	1.000	0.076
ABSLATIT	0.000	0.000	1.000	0.071
TROPICAR	-0.004	0.015	0.000	0.071
PI6090	0.000	0.000	1.000	0.062
CATH00	-0.003	0.014	0.000	0.058
SOCIALIST	-0.004	0.019	0.000	0.054
EUROPE	0.005	0.024	0.989	0.047
SPAIN	-0.002	0.010	0.006	0.033
IPRICE1	0.000	0.000	0.000	0.027
HERF00	-0.001	0.011	0.000	0.019
SCOUT	0.000	0.004	0.000	0.017
COLONY	-0.001	0.006	0.000	0.017
SQPI6090	0.000	0.000	0.981	0.017
ECORG	0.000	0.001	1.000	0.016
DENS60	0.000	0.000	0.000	0.015
PRIEXP70	-0.001	0.010	0.028	0.015
TROPPOP	-0.001	0.006	0.009	0.013
LT100CR	0.001	0.006	1.000	0.013
FERTLDC1	-0.001	0.009	0.000	0.013
CIV72	0.001	0.007	0.963	0.012
POP6560	0.008	0.094	0.980	0.012
BRIT	0.000	0.003	1.000	0.011
ZTROPICS	0.001	0.007	1.000	0.011
OIL	0.000	0.005	0.000	0.011
MUSLIM00	0.000	0.005	0.996	0.010
PROT00	0.000	0.005	0.988	0.010
ORTH00	-0.001	0.013	0.000	0.009
MINING	-0.001	0.015	0.000	0.009
DENS65C	0.000	0.000	0.981	0.008
POP1560	-0.001	0.022	0.028	0.008
GEEREC1	0.008	0.121	0.989	0.008
LANDAREA	0.000	0.000	0.274	0.008
SAFRICA	0.000	0.004	0.280	0.008
LANDLOCK	0.000	0.003	0.000	0.007
OPENDEC1	0.000	0.004	0.685	0.007
PRIGHTS	0.000	0.001	0.068	0.006
DPOP6090	-0.007	0.141	0.017	0.006
AVELF	0.000	0.003	0.894	0.005
GDPCH60L	0.000	0.002	0.832	0.005
NEWSTATE	0.000	0.001	0.654	0.005
OTHFRAC	0.000	0.002	0.143	0.005
TOT1DEC1	-0.001	0.020	0.005	0.005
TOTIND	0.000	0.005	0.579	0.005
MALFAL66	0.000	0.002	0.594	0.005
HINDU00	0.000	0.006	0.912	0.005
WARTIME	0.000	0.005	0.080	0.005
LHCPC	0.000	0.000	0.329	0.004
GDE1	0.001	0.027	0.932	0.004
AIRDIST	0.000	0.000	0.698	0.004
RERD	0.000	0.000	0.417	0.004
BUDDHA	0.000	0.004	0.167	0.004
DENS65I	0.000	0.000	0.571	0.004
WARTORN	0.000	0.001	0.186	0.004
ENGFRAC	0.000	0.002	0.414	0.004
GOVNOM1	0.000	0.009	0.556	0.004
GOVSH61	0.000	0.008	0.940	0.004
P60	0.000	0.004	0.436	0.004
GGCFD3	0.000	0.012	0.805	0.003
GVR61	0.000	0.007	0.620	0.003

Source: Author's construction, based on the AIPI and the database from Sala-i-Martin, Doppelhofer, and Miller (2004). The database is available here: https://www.openicpsr.org/openicpsr/project/116024/version/V1/view. Notes: The BMA considers a chain of three million recorded draws with one million burn-ins, by applying the birth-death sampler, using the BMS package from Zeugner and Feldkircher (2015). SD, and PIP stand for standard deviation, and posterior inclusion probability, respectively. Acronyms of covariates are available in Sala-i-Martin, Doppelhofer, and Miller (2004).

Table A.2: Descriptive statistics

	Economy	Region	Income level	ECI (multi., scaled)	AIPI (observed)	AIPI (predicted)	Overperformer
1	Afghanistan	South Asia	LIC	0.101	0.133	0.291	No
2	Albania	Europe and Central Asia	UMC	0.314	0.527	0.466	Yes (Local)
3	Algeria	Middle East and North Africa	UMC	0.106	0.370	0.358	No
4	Argentina	Latin America and Caribbean	UMC	0.574	0.474	0.600	No
5	Armenia	Europe and Central Asia	UMC	0.298	0.493	0.457	No
6	Australia	East Asia and Pacific	HIC	0.671	0.727	0.636	Yes (Global)
7	Austria	Europe and Central Asia	HIC	0.875	0.725	0.739	No
8	Azerbaijan	Europe and Central Asia	UMC	0.092	0.471	0.351	No
9	Bangladesh	South Asia	LMC	0.208	0.384	0.362	No
10	Belarus	Europe and Central Asia	UMC	0.348	0.471	0.483	No
11	Belgium	Europe and Central Asia	HIC	0.876	0.672	0.740	No
12	Benin	Sub-Saharan Africa	LMC	0.235	0.363	0.380	No
13	Bolivia	Latin America and Caribbean	LMC	0.173	0.377	0.339	No
14	Bosnia and Herzegovina	Europe and Central Asia	UMC	0.350	0.428	0.484	No
15	Botswana	Sub-Saharan Africa	UMC	0.161	0.413	0.387	No
16	Brazil	Latin America and Caribbean	UMC	0.638	0.501	0.633	Yes (Local)
17	Bulgaria	Europe and Central Asia	HIC	0.364	0.577	0.480	No
18	Burkina Faso	Sub-Saharan Africa	LIC	0.029	0.312	0.243	No
19	Cambodia	East Asia and Pacific	LMC	0.236	0.370	0.381	No
20	Cameroon	Sub-Saharan Africa	LMC	0.156	0.341	0.328	No
21	Canada	North America	HIC	0.877	0.713	0.740	No
22	Chile	Latin America and Caribbean	HIC	0.555	0.586	0.577	No
23	China	East Asia and Pacific	UMC	0.445	0.635	0.533	No
24	Colombia	Latin America and Caribbean	UMC	0.508	0.489	0.566	No
25	Congo, Dem. Rep.	Sub-Saharan Africa	LIC	0.106	0.247	0.294	No
26	Congo, Rep.	Sub-Saharan Africa	LMC	0.150	0.277	0.323	No
27	Costa Rica	Latin America and Caribbean	UMC	0.390	0.540	0.505	Yes (Local)
28	Croatia	Europe and Central Asia	HIC	0.558	0.582	0.578	No
29	Czechia	Europe and Central Asia	HIC	0.672	0.646	0.636	No
30	Denmark	Europe and Central Asia	HIC	0.819	0.779	0.711	Yes (Global)
31	Dominican Republic	Latin America and Caribbean	UMC	0.249	0.469	0.432	No
32	Ecuador	Latin America and Caribbean	UMC	0.201	0.442	0.407	No
33	Egypt, Arab Rep.	Middle East and North Africa	LMC	0.356	0.394	0.461	No
34	Ethiopia	Sub-Saharan Africa	LIC	0.293	0.254	0.419	No
35	Finland	Europe and Central Asia	HIC	0.856	0.758	0.730	Yes (Global)
36	France	Europe and Central Asia	HIC	0.862	0.698	0.733	No
37	Gabon	Sub-Saharan Africa	UMC	0.192	0.323	0.403	No
38	Georgia	Europe and Central Asia	UMC	0.540	0.530	0.582	Yes (Local)
39	Germany	Europe and Central Asia	HIC	0.932	0.753	0.768	No
40	Ghana	Sub-Saharan Africa	LMC	0.255	0.425	0.394	Yes (Local)
41	Greece	Europe and Central Asia	HIC	0.568	0.582	0.583	No
42	Guatemala	Latin America and Caribbean	UMC	0.437	0.390	0.529	No
43	Guinea	Sub-Saharan Africa	LMC	0.042	0.324	0.251	No
44	Honduras	Latin America and Caribbean	LMC	0.216	0.342	0.368	No
45	Hong Kong SAR, China	East Asia and Pacific	HIC	0.701	0.701	0.651	Yes (Global)
46	Hungary	Europe and Central Asia	HIC	0.719	0.563	0.660	No
47	India	South Asia	LMC	0.370	0.493	0.471	No
48	Indonesia	East Asia and Pacific	UMC	0.379	0.516	0.499	Yes (Local)
49	Iran, Islamic Rep.	Middle East and North Africa	UMC	0.439	0.378	0.530	No
50	Iraq	Middle East and North Africa	UMC	0.120	0.270	0.365	No

Table A.2: Descriptive statistics (cont'd)

	Country	Region	Income level	ECI (multi., scaled)	AIPI (observed)	AIPI (predicted)	Overperformer
51	Ireland	Europe and Central Asia	HIC	0.878	0.693	0.741	No
52	Israel	Middle East and North Africa	HIC	0.870	0.725	0.737	No
53	Italy	Europe and Central Asia	HIC	0.853	0.621	0.728	No
54	Côte d'Ivoire	Sub-Saharan Africa	LMC	0.117	0.366	0.302	No
55	Jamaica	Latin America and Caribbean	UMC	0.462	0.434	0.542	No
56	Japan	East Asia and Pacific	HIC	0.837	0.733	0.720	Yes (Global)
57	Jordan	Middle East and North Africa	UMC	0.368	0.483	0.493	No
58	Kazakhstan	Europe and Central Asia	UMC	0.173	0.552	0.392	Yes (Local)
59	Kenya	Sub-Saharan Africa	LMC	0.388	0.445	0.483	Yes (Local)
60	Kuwait	Middle East and North Africa	HIC	0.355	0.461	0.475	No
61	Kyrgyz Republic	Europe and Central Asia	LMC	0.414	0.426	0.500	Yes (Local)
62	Lao PDR	East Asia and Pacific	LMC	0.164	0.330	0.333	No
63	Lebanon	Middle East and North Africa	LMC	0.544	0.418	0.587	Yes (Local)
64	Libya	Middle East and North Africa	UMC	0.050	0.245	0.329	No
65	Lithuania	Europe and Central Asia	HIC	0.499	0.665	0.548	No
66	Madagascar	Sub-Saharan Africa	LIC	0.118	0.305	0.302	No
67	Malaysia	East Asia and Pacific	UMC	0.470	0.632	0.546	No
68	Mali	Sub-Saharan Africa	LIC	0.000	0.296	0.223	No
69	Mexico	Latin America and Caribbean	UMC	0.658	0.532	0.643	No
70	Moldova	Europe and Central Asia	UMC	0.311	0.481	0.464	No
71	Mongolia	East Asia and Pacific	UMC	0.102	0.484	0.356	No
72	Morocco	Middle East and North Africa	LMC	0.217	0.429	0.368	Yes (Local)
73	Mozambique	Sub-Saharan Africa	LIC	0.155	0.257	0.327	No
74	Myanmar	East Asia and Pacific	LMC	0.122	0.327	0.305	No
75	Namibia	Sub-Saharan Africa	UMC	0.241	0.420	0.428	No
76	Netherlands	Europe and Central Asia	HIC	0.890	0.766	0.747	Yes (Global)
77	New Zealand	East Asia and Pacific	HIC	0.710	0.754	0.655	Yes (Global)
78	Nigeria	Sub-Saharan Africa	LMC	0.161	0.336	0.331	No
79	North Macedonia	Europe and Central Asia	UMC	0.362	0.482	0.490	No
80	Norway	Europe and Central Asia	HIC	0.742	0.706	0.672	Yes (Global)
81	Oman	Middle East and North Africa	HIC	0.255	0.533	0.424	No
82	Pakistan	South Asia	LMC	0.220	0.369	0.370	No
83	Panama	Latin America and Caribbean	HIC	0.471	0.501	0.534	No
84	Papua New Guinea	East Asia and Pacific	LMC	0.056	0.290	0.261	No
85	Paraguay	Latin America and Caribbean	UMC	0.437	0.410	0.529	No
86	Peru	Latin America and Caribbean	UMC	0.311	0.491	0.464	No
87	Philippines	East Asia and Pacific	LMC	0.469	0.498	0.537	No
88	Poland	Europe and Central Asia	HIC	0.624	0.597	0.612	No
89	Portugal	Europe and Central Asia	HIC	0.617	0.646	0.608	No
90	Qatar	Middle East and North Africa	HIC	0.364	0.535	0.480	No
91	Romania	Europe and Central Asia	HIC	0.525	0.584	0.562	No
92	Russian Federation	Europe and Central Asia	HIC	0.456	0.559	0.527	No
93	Rwanda	Sub-Saharan Africa	LIC	0.246	0.437	0.388	Yes (Local)
94	Saudi Arabia	Middle East and North Africa	HIC	0.459	0.577	0.528	No
95	Senegal	Sub-Saharan Africa	LMC	0.228	0.396	0.376	No
96	Serbia	Europe and Central Asia	UMC	0.500	0.537	0.561	No
97	Singapore	East Asia and Pacific	HIC	0.718	0.801	0.660	Yes (Global)
98	Slovak Republic	Europe and Central Asia	HIC	0.528	0.592	0.563	No
99	Slovenia	Europe and Central Asia	HIC	0.627	0.634	0.613	No
100	South Africa	Sub-Saharan Africa	UMC	0.588	0.497	0.607	No

Table A.2: Descriptive statistics (end)

	Country	Region	Income level	ECI (multi., scaled)	AIPI (observed)	AIPI (predicted)	Overperformer
101	Korea, Rep.	East Asia and Pacific	HIC	0.670	0.727	0.635	Yes (Global)
102	Spain	Europe and Central Asia	HIC	0.776	0.648	0.689	No
103	Sri Lanka	South Asia	LMC	0.239	0.436	0.383	Yes (Local)
104	Sudan	Sub-Saharan Africa	LIC	0.041	0.233	0.251	No
105	Sweden	Europe and Central Asia	HIC	0.928	0.748	0.766	No
106	Switzerland	Europe and Central Asia	HIC	0.991	0.757	0.798	No
107	Taiwan, China	East Asia and Pacific	HIC	0.652	0.669	0.626	No
108	Tajikistan	Europe and Central Asia	LMC	0.058	0.366	0.262	No
109	Tanzania	Sub-Saharan Africa	LMC	0.249	0.352	0.390	No
110	Thailand	East Asia and Pacific	UMC	0.517	0.536	0.570	No
111	Togo	Sub-Saharan Africa	LIC	0.039	0.316	0.250	No
112	Tunisia	Middle East and North Africa	LMC	0.320	0.465	0.437	Yes (Local)
113	Türkiye	Europe and Central Asia	UMC	0.632	0.540	0.629	No
114	Uganda	Sub-Saharan Africa	LIC	0.309	0.354	0.430	No
115	Ukraine	Europe and Central Asia	UMC	0.356	0.512	0.487	Yes (Local)
116	United Arab Emirates	Middle East and North Africa	HIC	0.404	0.628	0.500	No
117	United Kingdom	Europe and Central Asia	HIC	0.980	0.731	0.792	No
118	United States	North America	HIC	1.000	0.771	0.803	No
119	Uruguay	Latin America and Caribbean	HIC	0.429	0.549	0.513	No
120	Uzbekistan	Europe and Central Asia	LMC	0.156	0.346	0.328	No
121	Venezuela, RB	Latin America and Caribbean	UMC	0.178	0.275	0.395	No
122	Viet Nam	East Asia and Pacific	LMC	0.285	0.482	0.414	Yes (Local)
123	Yemen, Rep.	Middle East and North Africa	LIC	0.002	0.253	0.225	No
124	Zambia	Sub-Saharan Africa	LMC	0.252	0.371	0.392	No
125	Zimbabwe	Sub-Saharan Africa	LMC	0.225	0.305	0.374	No

Source: The AIPI score for 2023 is obtained from the IMF website [https://www.imf.org/external/datamapper/datasets/AIPI] and the ECI on trade and research for 2021-22 are obtained from the OEC website [https://oec.world/en/rankings/eci/hs6/hs96?tab=ranking]. Notes: The first PCA of ECI trade and ECI research, with an eigenvalue exceeding 50 percent (normalized and scaled between 0 and 1 across the sample of countries) is reported here. Global and local overperformers are identified through a two-step approach: (i) comparing AIPI scores to the weighted median and average of income groups, and (ii) identifying scores exceeding predicted values based on economic complexity. Population size for 2023, used for the weighted analysis to obtain the AIPI predicted score, is sourced from the "wbopendata" module in Stata (Azevedo, 2020), with data for Taiwan Province of China obtained from the Statista website. For more details, see the ``Data and Methodology" and ``Empirical Findings" sections.